

$$\begin{aligned}
 2. P \rightarrow (P \wedge (Q \rightarrow P)) &\equiv P \rightarrow (P \wedge (\neg Q \vee P)) \text{ (Conditional-disjunction equivalence)} \\
 &\equiv P \rightarrow ((P \wedge \neg Q) \vee (P \wedge P)) \text{ (2nd Distributive Law)} \\
 &\equiv P \rightarrow ((P \wedge \neg Q) \vee P) \text{ (2nd Idempotent Law)} \\
 &\equiv P \rightarrow (P \vee (P \wedge \neg Q)) \text{ (2nd Commutative Law)} \\
 &\equiv P \rightarrow P \text{ (1st Absorption Law)} \\
 &\equiv \neg P \vee P \text{ (Conditional Disjunction Equivalence)} \\
 &\equiv P \vee \neg P \text{ (1st Commutative Law)} \\
 &\equiv T \text{ (1st Negation Law)}
 \end{aligned}$$

Principal Disjunctive Normal Form: $(\neg P \wedge \neg Q) \vee (\neg P \wedge Q) \vee (P \wedge \neg Q) \vee (P \wedge Q)$
 $\Sigma(0,1,2,3)$

$$\begin{aligned}
 3. (Q \rightarrow P) \wedge (\neg P \wedge Q) &\equiv (\neg Q \vee P) \wedge (\neg P \wedge Q) \text{ (Conditional-disjunction equivalence)} \\
 &\equiv (\neg P \wedge Q) \wedge (\neg Q \vee P) \text{ (1st Commutative Law)} \\
 &\equiv ((\neg P \wedge Q) \wedge \neg Q) \vee ((\neg P \wedge Q) \wedge P) \text{ (2nd Distributive Law)} \\
 &\equiv (\neg P \wedge (Q \wedge \neg Q)) \vee ((\neg P \wedge Q) \wedge P) \text{ (2nd Associative Law)} \\
 &\equiv (\neg P \wedge (Q \wedge \neg Q)) \vee (P \wedge (\neg P \wedge Q)) \text{ (2nd Commutative Law)} \\
 &\equiv (\neg P \wedge (Q \wedge \neg Q)) \vee (P \wedge \neg P) \wedge Q \text{ (2nd Associative Law)} \\
 &\equiv (\neg P \wedge F) \vee (F \wedge Q) \text{ (2nd Negation Law)} \\
 &\equiv (\neg P \wedge F) \vee (Q \wedge F) \text{ (2nd Commutative Law)} \\
 &\equiv F \vee F \text{ (2nd Domination Law)} \equiv F \text{ (2nd Identity Law)}
 \end{aligned}$$

Principal Conjunctive Normal Form: $(\neg P \vee Q) \wedge (\neg P \vee \neg Q) \wedge (P \vee \neg Q) \wedge (\neg P \vee \neg Q)$
 $\Pi(0,1,2,3)$

$$\begin{aligned}
 4. (\neg P \vee \neg Q) \rightarrow (P \rightarrow \neg Q) &\equiv (\neg P \vee \neg Q) \rightarrow (\neg P \wedge \neg(\neg Q)) \text{ (Section 1.83, Table 8, Equivalence)} \\
 &\equiv (\neg P \vee \neg Q) \rightarrow (\neg P \wedge \neg(\neg Q)) \text{ (Double Negation Law)} \\
 &\equiv (\neg P \vee \neg Q) \rightarrow ((P \wedge \neg Q) \vee (\neg P \wedge \neg Q)) \text{ (Conditional-disjunction equivalence)} \\
 &\equiv \neg(\neg P \vee \neg Q) \vee ((P \wedge \neg Q) \vee (\neg P \wedge \neg Q)) \text{ (1st De Morgan's Law)} \\
 &\equiv \neg(\neg(P \wedge Q)) \vee ((P \wedge \neg Q) \vee (\neg P \wedge \neg Q)) \text{ (Double Negation Law)} \\
 &\equiv (P \wedge Q) \vee ((P \wedge \neg Q) \vee (\neg P \wedge \neg Q)) \text{ (Double Negation Law)}
 \end{aligned}$$

PDF: $\Sigma(1,2,3)$